1. Revised objective

Giữ **inference pipeline objective**, tức là ở deploy time, caption vẫn được re-pool theo từng image summary q_i.

Vấn đề là train hiện tại chỉ tối ưu trên đường chéo:

$$T(c_i, q_i)$$

trong khi inference thực tế dùng:

$$T(c_j, q_i)$$

Vì không muốn enumerate mọi cặp (i, j) trong training, thay training scorer bằng một **amortized surrogate** có thể suy ra deployed text object từ một **prototype basis**.

Ý tưởng cốt lõi là:

- thay vì chỉ pool text dưới đúng q_i
- pool mỗi caption dưới từng prototype contextualized $\tilde{\theta}_n$
- dùng routing weights $\alpha_{i,n}$ của ảnh để tổng hợp lại text object gần với deployed object

---

2. Backbone and notation

Cho cặp huấn luyện thứ i, gồm ảnh x_i, caption c_i, và nhãn định danh y_i.

Visual encoder

$$v_i = f_v(x_i) \in \mathbb{R}^D \tag{1}$$

Text encoder

$$H_i = [h_{i,1}, h_{i,2}, \ldots, h_{i,L_i}] \in \mathbb{R}^{L_i \times D} \tag{2}$$

Prototype bank

$$\Theta_v = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_N \end{bmatrix} \in \mathbb{R}^{N \times D} \tag{3}$$

Contextualized prototype bank

$$\tilde{\Theta}_v = \mathrm{softmax}\left(\frac{\Theta_v \Theta_v^\top}{\sqrt{D}}\right)\Theta_v \in \mathbb{R}^{N\times D} \tag{4}$$

Mỗi hàng là một prototype đã được contextualize: $\tilde{\theta}_n$.

---

3. Image-conditioned prototype routing

Với ảnh x_i, tính routing distribution:

$$\alpha_{i,n} = \frac{\exp\left(v_i^\top \tilde{\theta}_n / \tau_p\right)}{\sum_{m=1}^{N} \exp\left(v_i^\top \tilde{\theta}_m / \tau_p\right)} \tag{5}$$

Sau đó xây dựng image-conditioned summary:

$$q_i = \sum_{n=1}^{N} \alpha_{i,n} \tilde{\theta}_n \in \mathbb{R}^D \tag{6}$$

Ý nghĩa:

- ảnh không chọn 1 prototype
- mà soft-route trên nhiều prototype
- tạo latent query q_i

---

4. Exact deployed text pooling

Inference thực tế dùng q_i để pool text.

Token scoring

$$s_{j|i,\ell}^{\text{exact}} = \frac{q_i^\top h_{j,\ell}}{\tau_t} \tag{7}$$

Attention weights

$$\beta_{j|i,\ell}^{\text{exact}} = \frac{\exp\left(s_{j|i,\ell}^{\text{exact}}\right)}{\sum_{r=1}^{L_j} \exp\left(s_{j|i,r}^{\text{exact}}\right)} \tag{8}$$

Pooled text

$$t_{j|i}^{\text{exact}} = \sum_{\ell=1}^{L_j} \beta_{j|i,\ell}^{\text{exact}} h_{j,\ell} \in \mathbb{R}^D \tag{9}$$

Đây là **deployed object thật**.

5. Prototype-basis text bank

Phần mới của method.

Thay vì pool theo q_i, pool theo từng prototype.

Token score theo prototype

$$u_{j,n,\ell} = \frac{\tilde{\theta}_n^\top h_{j,\ell}}{\tau_t} \tag{10}$$

Attention

$$\gamma_{j,n,\ell} = \frac{\exp(u_{j,n,\ell})}{\sum_{r=1}^{L_j} \exp(u_{j,n,r})} \tag{11}$$

Basis embedding

$$b_{j,n} = \sum_{\ell=1}^{L_j} \gamma_{j,n,\ell} h_{j,\ell} \in \mathbb{R}^D \tag{12}$$

Basis bank

$$B_j = \begin{bmatrix} b_{j,1} \\ b_{j,2} \\ \vdots \\ b_{j,N} \end{bmatrix} \in \mathbb{R}^{N \times D} \tag{13}$$

Ý nghĩa:

- mỗi caption có **N cách nhìn**
- mỗi cách tương ứng với một prototype

---

6. Amortized deployed text object

Reconstruct text object bằng routing của image:

$$\hat{t}_{j|i} = \sum_{n=1}^{N} \alpha_{i,n}\, b_{j,n} \in \mathbb{R}^D \tag{14}$$

$$\hat{t}_{j|i} \approx t_{j|i}^{\text{exact}} \tag{15}$$

Với diagonal:

$$\hat{t}_{i|i} = \sum_{n=1}^{N} \alpha_{i,n} b_{i,n} \tag{16}$$

Ý nghĩa:

- ảnh cung cấp α_i
- text cung cấp B_j
- không cần pairwise pooling

---

7. Projection heads

$$z_i^v = g_v(v_i) \in \mathbb{R}^{D_o} \tag{17}$$

$$\hat{z}_{j|i}^t = g_t(\hat{t}_{j|i}) \tag{18}$$

$$z_{i|i}^{t,\text{exact}} = g_t(t_{i|i}^{\text{exact}}) \tag{19}$$

---

8. Training pipeline

Diagonal fidelity loss

Ép surrogate bám theo deployed object:

$$\mathcal{L}_{\text{diag}} = \frac{1}{B} \sum_{i=1}^{B} \left(1 - \cos\left(\hat{z}_{i|i}^t, \text{sg}[z_{i|i}^{t,\text{exact}}]\right)\right) \tag{20}$$

Proxy loss

Không dùng in-batch negatives.

Image branch

$$\mathcal{L}_{\text{proxy}}^v = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(\cos(z_i^v, p_{y_i})/\tau_c)}{\sum_{c=1}^{C} \exp(\cos(z_i^v, p_c)/\tau_c)} \tag{21}$$

Text branch

$$\mathcal{L}_{\text{proxy}}^t = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(\cos(\hat{z}_{i|i}^t, p_{y_i})/\tau_c)}{\sum_{c=1}^{C} \exp(\cos(\hat{z}_{i|i}^t, p_c)/\tau_c)} \tag{22}$$

$$\mathcal{L}_{\text{proxy}} = \mathcal{L}_{\text{proxy}}^v + \mathcal{L}_{\text{proxy}}^t \tag{23}$$

Alignment loss

$$\mathcal{L}_{\text{align}} = \frac{1}{B} \sum_{i=1}^{B} \left(1 - \cos(z_i^v, \hat{z}_{i|i}^t)\right) \tag{24}$$

Regularization

$$\mathcal{L}_{\text{div}} = \left\| \frac{\Theta_v \Theta_v^\top}{\|\Theta_v\|_F^2} - I \right\|_F^2 \tag{25}$$

$$\mathcal{L}_{\text{bal}} = \text{BalancingRegularizer}(\alpha) \tag{26}$$

Total loss

$$\mathcal{L} = \lambda_{\text{proxy}} \mathcal{L}_{\text{proxy}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{diag}} \mathcal{L}_{\text{diag}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} \tag{27}$$

---

9. Inference pipeline

Precompute gallery

- v_i
- α_i
- q_i

Query-time

$$t_{j|i}^{\text{exact}} = T(c_j, q_i)$$

$$z_{j|i}^t = g_t(t_{j|i}^{\text{exact}})$$

$$s(i,j) = \cos\left(z_i^v, z_{j|i}^t\right) \tag{28}$$

---

10. Optional amortized scorer

$$\hat{s}(i,j) = \cos\left(z_i^v, g_t(\hat{t}_{j|i})\right) \tag{29}$$

---

11. Complexity

- Exact pairwise: $O(B^2 L)$
- Proposed:
  - basis pooling: $O(BNL)$

- routing: $O(BN)$
- diagonal exact: $O(BL)$

Vì $N \ll B$, chi phí giảm mạnh.

---

12. Vì sao giải quyết vấn đề

Issue: train và inference dùng **khác text object**

Pipeline mới:

1. Inference giữ nguyên $T(c_j, q_i)$
2. Training học:

$$\hat{t}_{j|i} = \sum_n \alpha_{i,n} b_{j,n}$$

3. Dùng $\mathcal{L}_{\text{diag}}$ để neo về deployed operator
4. Không cần pairwise enumeration

⇒ Training học **amortized deployed operator**

---

13. Tóm tắt

- image → α_i, q_i
- text → B_j
- training dùng $\hat{t}_{j|i}$
- diagonal distill từ $T(c_i, q_i)$
- inference giữ nguyên